

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

FINAL SEMESTER EXAMINATION OF 1st SEMESTER 1st YEAR OF BATCH
2025/FALL ARTIFICIAL INTELLIGENCE OF 2026

SUBJECT: FEHM UL QURAN (THEORY)

DATED: 12-06-2026 TIME ALLOWED 1hr

MAX. MARKS 25

Student Name: Syed Shehryar Tariq

Student Id Number: 25F-01-077

Instructions:

1. This paper contains 05 questions.
2. Use of Calculator is allowed Not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

(CLO, CLO, MARKS:5)

Question No. 1: Write the Urdu meanings of the following words.

سوال نمبر 1: درج ذیل الفاظ کے اردو معنی لکھیں۔

هَذَا وَلَعِبًا / ایک صراف اور کھیل

هَذَا ذَلِك الْخِزْيُ / وہ رسوا

طُغْيَانًا وَكُفْرًا

هَذِهِ نَافَةٌ

رَجَسٌ وَغَضَبٌ / کوئی عیب اور غضب

أَرْكَهَ هُوَ / پاکیزہ

هَذَا بَلْعَةٌ

إِنَّ السَّاعَةَ

هَذَا مِلْحٌ

لِلْخَرْبِ

(CLO, CLO, MARKS: 5)

Question No. 2: Explain the meanings of the following words in Arabic.

سوال نمبر 2: درج ذیل الفاظ کے عربی معنی بیان کریں۔

ناشکری وزن

قرب تر

نصیحت افک

شہر مذاق

گھر تنگی

(CLO, MARKS:5)

Question No. 3: Add the correct diacritical marks (A'rāb) to the following words.

بسم الله الرحمن الرحيم

لسان - لِسَانٌ

مَنَاعٌ

نَحَارَةٌ

فَرِيقٌ

رِزْقٌ

حَسَنَةٌ

حَسَنَةٌ

الْفَضَّةُ

مَرَضٌ

سَيِّدَةٌ

سَيِّدَةٌ

الْفُسُوقُ

(PLO, MARKS:5)

Question No. 4: Explain the meanings/concepts of the following terms in two lines.

بسم الله الرحمن الرحيم

1. رسول: ایک رسول / اللہ کا پیغام انسانوں تک پہنچانے والا

2. رشک:

3. الصفاء و المروءة: حرم میں موجود دو پہاڑ جنہیں مسلمان حج

میں چڑھتے اترتے ہیں، عفاہ اور مروءہ

4. الصلاة: نماز / اللہ

5. الحرب/ القتال:

6. الذب:

7. الدين:

سوال نمبر 3: درج ذیل الفاظ پر

8. الانجیل :

9. الصوم:

10. الغیب:

(PLO, MARKS:5)

Question No. 5: Explain the meanings or concepts of the following sentences.

سوال نمبر 5: درج ذیلوں کے مفہوم یا مطلب واضح کیجیے۔

1. وَالَّذِينَ هُمْ لِلزَّكَاةِ فَاعِلُونَ

2. وَأَنْ لَّيْسَ لِلْإِنْسَانِ نُفْسَانِ إِلَّا نَفْسٌ مَّا سَعَى

3. وَفِي أَمْوَالِهِمْ خَيْرٌ لِّلسَّائِلِ وَالنَّازِغِ رُوم 4. ا

‘يَبْتَغِي نَفْسًا لِلرَّحْمَنِ صَوْمًا

5. إِنَّ الشَّيْطَانَ لِرَبِّهِ الْإِنْسَانِ سَاحِقٌ

THE END



DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

Name of Muhammad Farshad

FINAL EXAMINATION OF 2nd SEMESTER, 1st YEAR OF BS Artificial Intelligence,
SECTIONS (A1, A2, A3), OF 2026

SUBJECT: COMMUNICATION SKILLS(THEORY)

DATED: 16-06-2026

TIME ALLOWED 02 HOURS

MAX. MARKS 25

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 03 questions.
2. Use of Calculator is allowed/Not allowed.
3. Cheating of any type will disqualify the candidate.
4. Do not detach the sheets. (Paper will be cancelled, if the sheets are detached).
5. Write your answers in ink. For drawing pencils may be used.
6. The work must be neat & clean.
7. Exchange / Borrow of Calculator or Stationery is not allowed.
8. Mobile Phones must be powered off and submitted at the given counters during the exam.
9. The Answer Script must be returned to invigilator before leaving exam hall.

NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO. 1 Explain the Barriers in Communication. Classify and explain the major barriers in communication, including Physical/Environmental Barriers, Physiological Barriers, Psychological/Perceptual Barriers, Cultural Barriers, Linguistic Barriers, Emotional Barriers, and Gender Barriers. Support your answer with relevant examples illustrating their impact on the communication process.

(C-2, CLO-1, PLO-09, MARKS-9)

Q.NO. 2 Discuss the Communication Fundamentals in detail. Elaborate on the significance of Empathy, Conversational Skills, Active Listening, Respectful Vocabulary, Introspection, and Turn-Taking in developing effective interpersonal and professional communication.

(C-2, CLO-1, PLO-09, MARKS-8)

Q.NO. 3 Write a detailed note on Letter, Writing. Compare and contrast the purpose, structure, similarities, and differences of the following types of letters: Sales Letters, Inquiry Letters, Complaint Letters, Cover Letters, Good News Letters, Bad News Letters, Formal Letters, and Informal Letters.

(C-3, CLO-2, PLO-09, MARKS-8)

THE END



Name of Subject Teacher: Faisal

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

**FINAL EXAMINATION OF 2ND SEMESTER, 1ST YEAR (FALL SEMESTER 2025) OF
BS ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2, & A3), OF 2026**

SUBJECT: (PAKISTAN STUDIES)

DATED: 10-06-2026

TIME ALLOWED 120 MINUTES

MAX. MARKS 25

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 03 questions.
2. Use of Calculator is allowed/Not allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

(CLO-1, PLO-6, 5 MARKS)

Q.NO. 1 (A) "Geography is destiny." Discuss this statement with special reference to Pakistan's location and physical features, including opportunities and constraints in the 21st century.

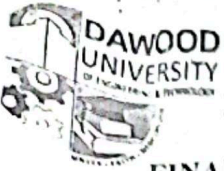
(CLO-2, PLO-6, 10 MARKS)

Q.NO. 2 (A) Write a comprehensive essay on the evolution of the Kashmir conflict from 1947 to 2026, highlighting major turning points and explaining why it remains unresolved.

(CLO-2, PLO-6, 10 MARKS)

Q.NO. 3 (A) The 1956 Constitution was a Westminster-style parliamentary system, the 1962 was a centralized presidential system, while the 1973 attempted a balanced parliamentary federal system." Critically compare the three constitutions with reference to their salient features, strengths, and weaknesses.

THE END



FINAL EXAMINATION OF 2ND SEMESTER, 1ST YEAR 2025/FALL BATCH OF BS
ARTIFICIAL INTELLIGENCE SECTIONS (A1, A2 & A3) OF 2026

SUBJECT: DIGITAL LOGIC DESIGN (THEORY)

DATED: 08-06-2026 TIME ALLOWED 02 HOURS MAX. MARKS 25

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains 05 questions.
2. Use of Calculator is allowed/Not allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO. 1 (A) Write the simplified Boolean expression of following expression using K map.
 $X_1 = A'BC + A'B'C + ABC' + ABC + AB'C$ (CLO-2, 3 marks)

(B) Simplify the following function using K-map in POS form.
 $X_2 = B'D + B'C + ABCD$ (CLO-2, 4 marks)
 $X_3 = A'BD + ABC'D'$

Q.NO. 2 (A) Design a circuit that can perform both addition and subtraction based on a control signal M. Explain the function of the mode control input M. (CLO-2, 5 marks)

(B) Find the addition and subtraction output when $A = 1011_2$ and $B = 1101_2$. (CLO-2, 5 marks)

Q.NO. 3 (A) Explain the working of a NOR-based SR Flip-Flop and a JK Flip-Flop. Include the logic diagrams, circuit symbols, and truth tables for both. (CLO-1, 5 marks)

(B) Differentiate between a Latch and a Flip-Flop by outlining at least three key distinctions. (CLO-1, 3 marks)

THE END

**FINAL SEMESTER EXAMINATION OF 2ND SEMESTER 1ST YEAR OF BATCH-
2025/FALL BS ARTIFICIAL INTELLIGENCE OF SPRING 2026**

SUBJECT: OBJECT ORIENTED PROGRAMMING (THEORY)

DATED: 02-06-2026

TIME ALLOWED 03 Hours

MAX. MARKS 50

Student Name: _____

Student Id Number: _____

Instructions:

1. This paper contains **05** questions.
2. Use of Calculator is allowed.
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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO.1 (CLO-1 C2 GA2)

[Marks 10]

a) [4 Marks]

Distinguish between early binding and late binding with suitable examples.

b) [6 Marks]

Review a C++ program which uses a base-class pointer to demonstrate runtime polymorphism through a virtual function showData(). Your program should:

- Define Employee class with a virtual showData(), and Manager and Engineer classes overriding it.
- Use pointer of base-class type to call showData() on each object.
- Explain the output: state which version of showData() is called and why.

[Marks 10]

Q.NO.2 (CLO-3 C3 GA4)

a) [3 Marks]

Determine why is an object of an abstract class not allowed to be instantiated? Explain with an example.

b) [7 Marks]

Implement a class hierarchy consisting of an abstract class Shape containing a pure virtual function area(). Derive classes Rectangle and Circle, each providing a concrete implementation of area(). Demonstrate polymorphic behavior using a base-class pointer in main(). Show sample output and explain which area() is invoked at each call and why.

[Marks 10]

Q.NO.3 (CLO-3 C3 GA4)

a) [4 Marks]

Determine the concept of Multilevel Inheritance. Discuss the order of execution of constructors and destructors in a multilevel inheritance hierarchy.

b) [6 Marks]

Implement a C++ program using the above hierarchy with parameterized constructors in each class (Person, Employee, Manager). Each constructor should print a message such as "Person constructor called" to verify invocation order at runtime. Show the expected output and trace the constructor/destructor sequence.

Q.NO. 4 (CLO-2 C3 GA3)

[Marks 10]

a) [3 Marks]

Conduct a comparison of the role, advantages and limitations of friend functions and friend classes in C++.

b) [7 Marks]

Use a class Distance containing two data members: feet (int) and inches (int) to overload the + operator using a friend function to add two Distance objects. Handle the case where inches ≥ 12 by carrying over to feet.

Write a complete program with main() demonstrating the addition.

Q.NO. 5 (CLO-1 C2 GA2)

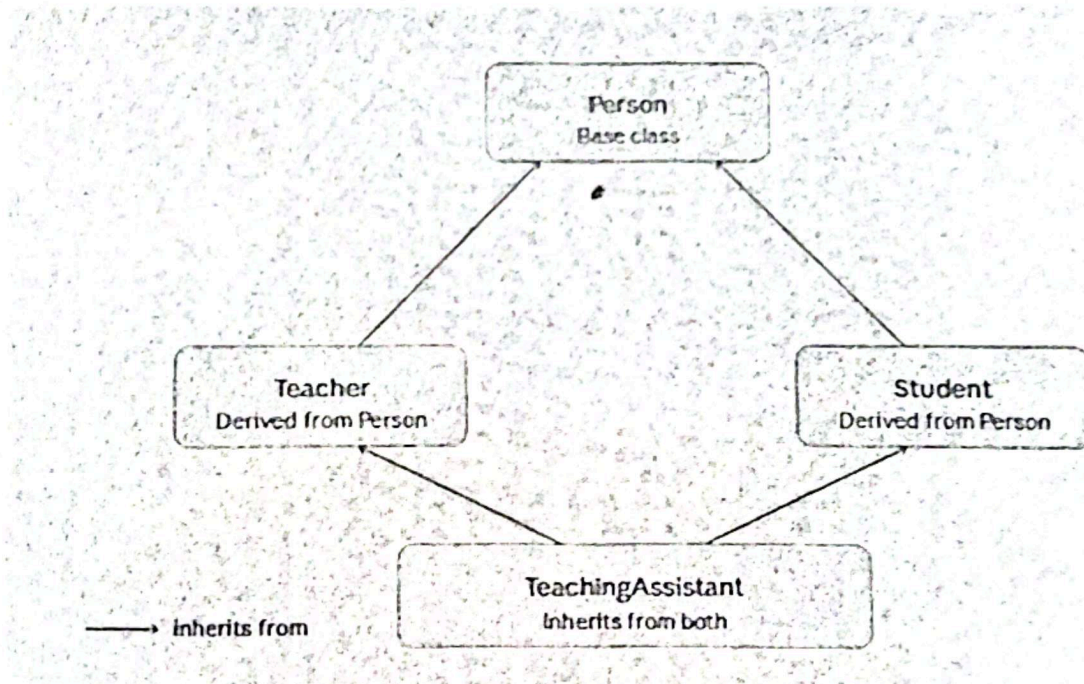
[Marks 10]

a) [4 Marks]

Explain the Diamond Problem in Multiple Inheritance with the help of a suitable class hierarchy diagram. Discuss the ambiguity that arises when accessing inherited members.

b) [6 Marks]

Describe how does Virtual Inheritance resolve the Diamond Problem? Write a C++ program implementing the following hierarchy and explain the output:



THE END

DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY, KARACHI

FINAL EXAMINATION OF 2nd SEMESTER, 1st YEAR (FALL SEMESTER 2025) OF BS
ARTIFICIAL INTELLIGENCE SEC (A, B & C) OF 2026

SUBJECT: LINEAR ALGEBRA (THEORY)

DATED: 04-06-2026

TIME ALLOWED 03 HOURS

MAX. MARKS 50

Student Name: _____

Student Id Number: _____

Instructions:

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NOTE: - ATTEMPT ALL THE FOLLOWING QUESTIONS.

Q.NO.1 Illustrate the inverse of matrix and rank of matrix

$$\begin{bmatrix} 2 & -3 & 1 \\ 2 & 0 & -1 \\ 1 & 4 & 5 \end{bmatrix}$$

(CLO-1, GA-2, C2, 10 MARKS)

Q.NO.2(a) Solve the following linear equations by using Gauss Jordan Method

$$\begin{aligned} 10x + y + z &: 12 \\ 2x + 10y + z &: 13 \\ x + y + 5z &: 7 \end{aligned}$$

(CLO-2, GA-3, C3, 10 MARKS)

Q.NO.2(b) Solve the following system of equations by using Gauss Elimination Method

$$\begin{aligned} x + 4y - z &: -5 \\ x + y - 6z &: -12 \\ 3x - y - z &: 4 \end{aligned}$$

(CLO-3, GA-3, C3, 10 MARKS)

Q.NO.3 Apply Eigen values and Eigen vectors of the following matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ 6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

(CLO-3, GA-1, C3, 10 MARKS)

Q.NO.4(a) Show that the set of Q^+ of all rational number forms an abelian group under the operation defined by $a * b = ab/2$.

Q.NO.4(b) Show that the set $G = \{1, 2, 3, 4, \dots\}$ is an abelian group.

(CLO-3, GA-1, C3, 5 MARKS)

Q.NO.5 Show that the set of rational numbers Q is a ring under the compositions $\oplus$ and $\odot$ defined as $a \oplus b = a + b - 1$ and $a \odot b = a + b - ab$, $\forall a, b \in Q$.

(CLO-3, GA-1, C3, 5 MARKS)

THE END